

R2: Spectral Shape

9 October 2026 (rerun)

Where should the wave energy lie for the inverse to work best?

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SHARC Code: `runR2SpectralShape.m`, `buildR2Seas.m`, commit c09a18e Results: `Results/R2/R2_20261009_100148`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 3 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). Settings unchanged; tester 10/10. This summary supersedes R2_20261003_130557. Precision improved six- to eightfold for every sea, so every sea is now limited by the Welch bias rather than by precision. As a result the M1 stress sea changed from $\omega_p = 3.5$, $\gamma_J = 7$ to $\omega_p = 6$, $\gamma_J = 7$.

Summary

Question. The floe (p_0), layout (Level 2C, reference s26), noise (LSM6DSV16X), record ($N = 65536$) and the wave energy inside the retained band are all fixed. How does the distribution of that energy across frequency change precision and Welch bias? There are 16 pre-registered seas: JONSWAP with $\omega_p \in \{3.5, 4.25, 5, 6, 7.5\}$ rad/s and $\gamma_J \in \{1, 3.3, 7\}$, plus one bimodal sea. Each is evaluated at $L = 1024, 2048$ and 4096 .

Outcome. The exact transmissibility and its Jacobian are identical for every sea (difference 0), yet the practical inverse depends on the sea. At $L = 2048$, $\sigma_{\ln \beta}$ ranges from 0.025 to 0.091% (a factor 3.6 at equal in-band energy) and d_W from 0.370 to 0.558. **15 of 16 seas are Identifiable at $L = 2048$.** The exception is $\omega_p = 6$ rad/s, $\gamma_J = 7$, which is Marginal on bias ($d_W = 0.558$) and becomes Identifiable at $L = 4096$. $L = 1024$ is Not identifiable for every sea, and $L = 4096$ is Identifiable for all 16 and gives the smallest RMSE in $\ln \beta$ for all 16. Precision and bias pull in opposite directions: energy in the upper band gives the best precision, while a narrow peak at about 6 rad/s gives the largest bias. M1 case (by the pre-fixed rule): $\omega_p = 6$ rad/s, $\gamma_J = 7$, $q = 1.116$, bias-limited.

1 What the study does

Controlled experiment. Only the sea and L change. The following are fixed:

- $p = p_0$, Level 2C with reference s26, LSM6DSV16X noise;
- $N = 65536$, $\Delta t = 0.1$ s, the retained band 3.178 to 8.301 rad/s;
- the cached EMM field, the estimator and the class thresholds.

These were checked by the tester (10/10), and the reference sea reproduces `evaluateScenario` bitwise: $\sigma = [0.0499\%, 0.0083\%]$, $d_W = 0.4355$, $\kappa = 83.2$, 84 bins. The bimodal sea has peaks at 3.5 and 6.5 rad/s ($\gamma_J = 7$ and 3.3), carrying 40 and 60% of the in-band variance.

Fixed in-band variance (a logged clarification of the programme’s “fixed H_s ”). Every sea satisfies

$$\int_{3.178}^{8.301} S_{\eta}^{(q)}(\omega) d\omega = \int_{3.178}^{8.301} S_{\eta, \text{ref}}(\omega) d\omega = 1.407 \times 10^{-4} \text{ m}^2$$

(spread 5.6×10^{-16} relative). R2 therefore changes *where* the usable energy lies, not *how much* there is; R1 covers the level. The single-peak seas have H_s from 0.049 to 0.072 m, and the reference keeps exactly $H_s = 0.05$ m.

Why the shape can matter at all. For one incident component, $S_{jr} = H_j H_r^* S_{uu}$ and $S_{rr} = |H_r|^2 S_{uu}$, so $T_j = S_{jr}/S_{rr} = H_j/H_r$. The exact transmissibility and its Jacobian J therefore do not depend on S_{uu} . Two practical quantities do:

- The covariance Σ_k depends on the per-bin signal-to-noise ratio, so the sea decides which bins carry information in $F = \sum_k J_k^T \Sigma_k^{-1} J_k$.
- The Welch estimate smooths before taking the ratio, $\hat{T}_j \approx (W * S_{jr}) / (W * S_{rr})$. Where T varies across the window W , the result depends on how S_{uu} weights the neighbouring frequencies, so the Welch bias depends on the sea.

The feature bias b_W in (ii) is a ratio effect and does not depend on the wave level. Its significance $d_W = \sqrt{\delta \theta_W^T F \delta \theta_W}$ does, because it is measured against F .

2 Results

2.1 Production setting, $L = 2048$

ω_p (rad/s)	$\sigma_{\ln \beta}$			d_W		
	$\gamma_J = 1$	3.3	7	$\gamma_J = 1$	3.3	7
3.5	0.064%	0.077%	0.091%	0.426	0.398	0.370
4.25	0.052%	0.061%	0.071%	0.423	0.413	0.404
5	0.043%	0.050% [†]	0.057%	0.410	0.436 [†]	0.460
6	0.034%	0.040%	0.045%	0.394	0.483	0.558 (M)
7.5	0.027%	0.025%	0.025%	0.447	0.473	0.491
bimodal 3.5 + 6.5	0.041%	$(\sigma_{\ln R} 0.006\%, \kappa 87.6)$		0.455		

[†]Reference sea (= E1/ p_0). $\sigma_{\ln R}$ follows $\sigma_{\ln \beta}$ (0.004 to 0.017%); $\kappa(F)$ 70 to 106. All Identifiable except (M) Marginal. Pre-amendment: $\sigma_{\ln \beta}$ 0.161 to 0.766%, d_W 0.048 to 0.359, all 16 Identifiable.

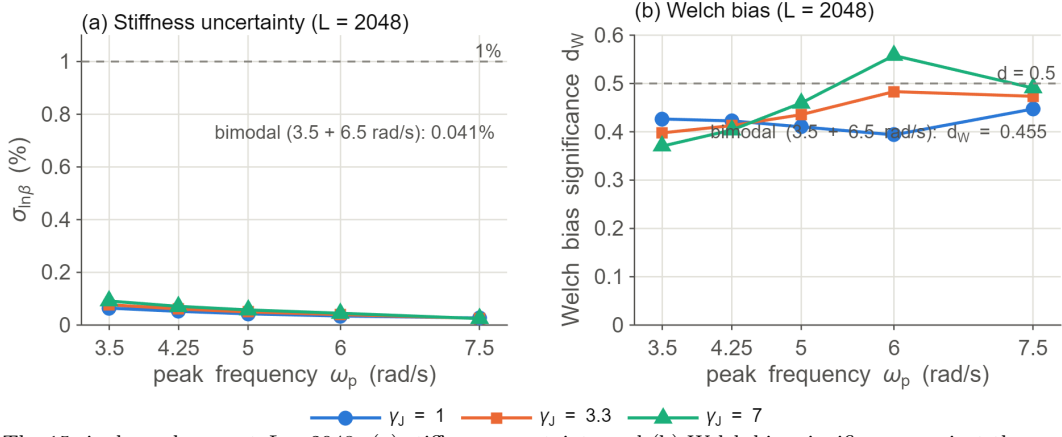


Figure 1. The 15 single-peak seas at $L = 2048$: (a) stiffness uncertainty and (b) Welch bias significance against the peak frequency, one line per peakedness. The bimodal sea has no single peak frequency and is annotated. Dashed: the Identifiable limits.

Peak location dominates the precision. For every γ_J , moving the peak from 3.5 to 7.5 rad/s lowers $\sigma_{\ln \beta}$ monotonically: by a factor 2.4 for $\gamma_J = 1$, 3.1 for 3.3 and 3.6 for 7.

Peakedness magnifies placement. At $\omega_p = 3.5$ a narrower sea is worse (0.064, 0.077, 0.091% for $\gamma_J = 1, 3.3, 7$). At 5 rad/s the same holds but less strongly (0.043 to 0.057%). At 7.5 rad/s the order reverses slightly (0.027, 0.025, 0.025%). Every $\sigma_{\ln \beta}$ is at least 11 times below the 1% limit, so precision no longer decides any class.

The bias decides, and it moves the other way.

- For the broad sea ($\gamma_J = 1$), d_W is nearly flat (0.39 to 0.45).
- For the narrower seas it rises with ω_p to a maximum at 6 rad/s: 0.483 for $\gamma_J = 3.3$, and 0.558 for $\gamma_J = 7$, the one Marginal case.
- The low-frequency narrow sea, which is least precise, is the least biased (0.370).

Before the amendment the d_W maximum sat at 5 rad/s and the high-frequency seas were almost bias-free ($d_W \approx 0.05$). The feature bias b_W has not changed. What changed is F : the amendment raised the signal most at high frequency ($\alpha \propto \omega^2$), so Σ_k fell most there, and the same feature bias now counts for more. d_W at $\omega_p = 7.5$, $\gamma_J = 7$ rose tenfold (0.048 to 0.491), against 1.4 times for the reference.

The bimodal sea is better than the reference in precision (0.041% against 0.050%) and slightly worse in bias (0.455 against 0.436).

2.2 Exact inverse versus finite estimator

Across all 16 seas $\max_q |f_0^{(q)} - f_{0,\text{ref}}| = 0$ and $\max_q |J^{(q)} - J_{\text{ref}}| = 0$: the physical inverse map does not change at all. The Welch feature bias, by contrast, does: $\|b_W - b_{W,\text{ref}}\|/\|b_{W,\text{ref}}\|$ ranges from 0.20 to 556. Everything in Figure 1 comes from the experiment and the estimator.

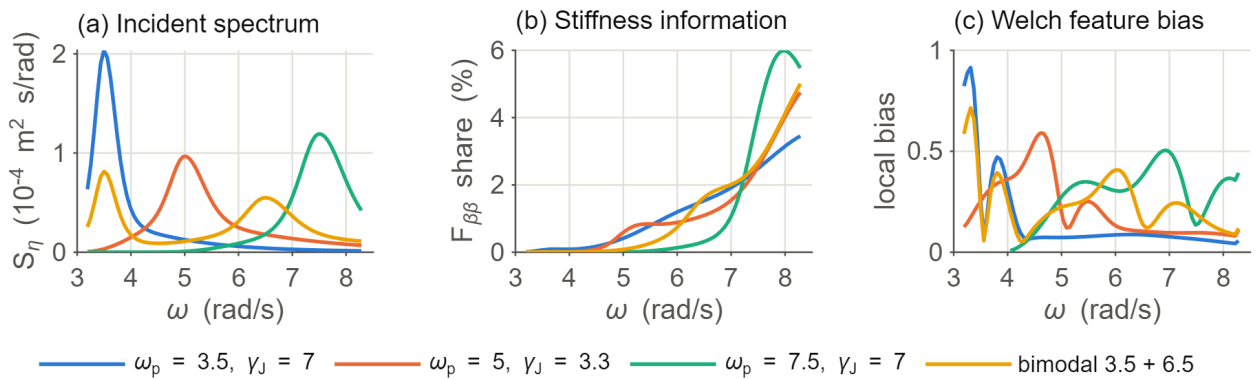


Figure 3. Mechanism at $L = 2048$ for four seas of equal in-band variance. (a) Incident spectrum on the retained bins. (b) Share of the diagonal Fisher information $F_{\beta\beta,k}$ per accepted bin. This is a frequency-local diagnostic, not a decomposition of $\sigma_{\ln \beta}$, which also depends on $F_{\beta R}$ and F_{RR} . (c) Local feature-bias significance $\sqrt{b_{W,k}^T \Sigma_k^{-1} b_{W,k}}$ per bin. This is also a diagnostic, not a decomposition of d_W : the global bias is projected through F , and bin contributions can cancel.

The upper band carries the stiffness information. The share of $F_{\beta\beta}$ from accepted bins above 6 rad/s is:

sea	$\omega_p = 3.5, \gamma_J = 7$	reference	$\omega_p = 7.5, \gamma_J = 7$	bimodal
share above 6 rad/s	83%	84%	99%	95%

Even the sea with its peak at 3.5 rad/s draws 83% of its stiffness information from its weak tail above 6 rad/s. Energy near the lower band edge is spent in bins that say little about stiffness, so moving the same energy upward lowers $\sigma_{\ln \beta}$.

The same reweighting plausibly explains the bias maximum. A narrow peak at about 6 rad/s puts the steep spectral flanks, where the window bias is formed, onto the bins that now carry most of $F_{\beta\beta}$. A peak at 7.5 rad/s is partly cut by the upper band edge. A peak at 3.5 rad/s places its flanks where F is small. Only the global d_W classifies.

2.3 Welch length

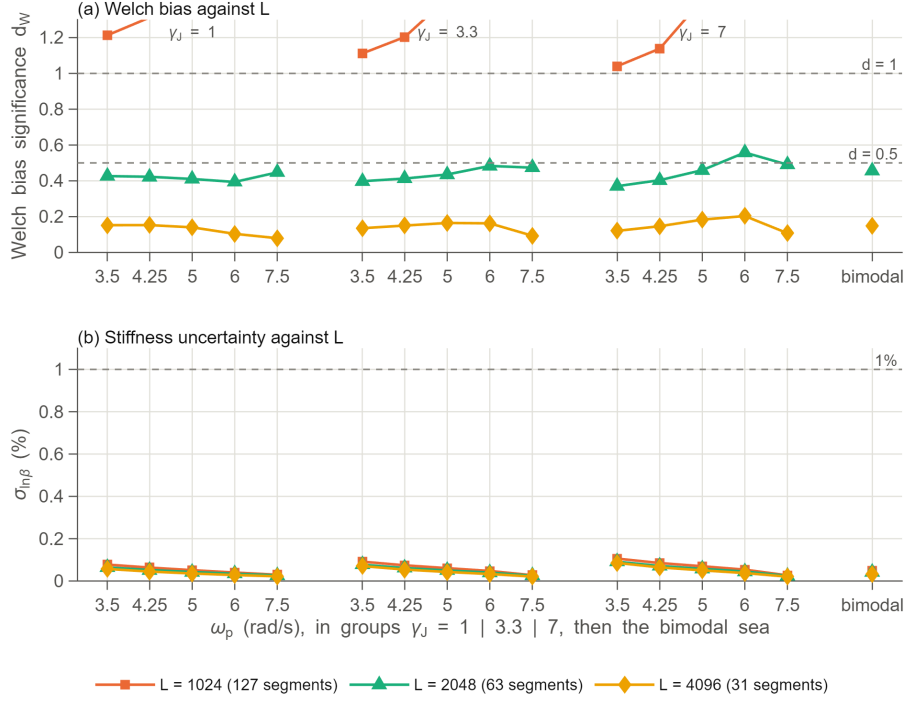


Figure 2. All 16 seas at $L = 1024, 2048, 4096$ ($N = 65536$: 127, 63, 31 segments): (a) Welch bias significance, (b) stiffness uncertainty. Colours as in D2.

L	Identifiable	Marginal	Not identifiable	d_W range	pre-amendment
1024	0	0	16	1.04 to 2.07	3 I, 12 M, 1 N
2048	15	1 ($\omega_p = 6, \gamma_J = 7$)	0	0.37 to 0.56	16 I
4096	16	0	0	0.08 to 0.20	16 I

$\sigma_{\ln\beta}$ barely depends on L (Figure 2b; reference 0.061, 0.050, 0.043%). The bias significance falls by roughly a factor 3 per doubling of L , as in D2. Using the RMSE $\sqrt{\sigma_{\ln\beta}^2 + \delta\theta_{W,\beta}^2}$, $L = 4096$ is best for all 16 seas at $N = 65536$.

3 Interpretation

- Where the energy lies, not how much, still sets the precision.** Equal in-band variance does not mean equal information: redistributing the same energy changes $\sigma_{\ln\beta}$ by a factor 3.6. Since every $\sigma_{\ln\beta}$ is below 0.1%, this no longer changes any class.
- The sea's shape now matters through the bias.** Every sea is bias-limited at $L = 2048$. The class is set by how the spectral flanks line up with the informative upper band, not by how much information the sea supplies. This is the same picture as D2, R1 and R4 after the amendment: precision is not limiting; fixed biases are.
- Precision and bias favour different seas.** Energy in the upper band gives the best precision. A narrow peak at about 6 rad/s gives the largest bias. A narrow sea at the lower edge gives the worst precision but the smallest bias. There is no single best sea at a fixed L ; a longer L removes the conflict.
- The upper part of the band is more informative for this configuration.** This holds for p_0 , Level 2C and the basin band. It is not a general claim that higher frequencies are always better: A1 showed that kR , which changes with the floe, organises the precision.
- The physics is sea-independent; the experiment and the estimator are not.** T and J are unchanged; F , σ and d_W change. This separates the identifiability of the forward model from the information a given sea supplies and the bias a given estimator adds.
- $L = 2048$ **no longer passes the robustness test cleanly.** 15 of 16 seas are Identifiable, and the exception is Marginal by a small margin ($d_W = 0.558$ against 0.5). $L = 1024$ fails for every sea. $L = 4096$ passes for all 16 seas with $d_W \leq 0.20$ and is the RMSE optimum, in line with D2 (RMSE-optimal from 60 min). The $L = 2048$ baseline is retained for comparability; $L = 4096$ is the recommended production setting for a full 109 min record.
- M1 stress case:** $\omega_p = 6, \gamma_J = 7, q = 1.116$, with $d_W/0.5$ binding ($\sigma_{\ln\beta}/1\% = 0.045$). The previous case, $\omega_p = 3.5, \gamma_J = 7$, now has $q = 0.74$. M1 therefore probes whether realised records reproduce the predicted Welch bias, not weak Fisher information. This is the sharper test for the post-amendment picture, because the bias is now what decides the class. The change follows the pre-registered rule and is logged as a re-resolution (2026-10-09).

Limitations.

- One floe and one layout (p_0 , Level 2C), not the A1 domain.
- Fixed in-band variance rather than fixed H_s , deliberately (H_s varies between seas).
- One noise case; R1 and R3 cover the noise.
- The JONSWAP family plus one bimodal sea, and one incident direction (V2, D3).
- Analytic expectations, not realised records; M1 tests the selected sea.
- Model truncation bias is not included here (V1).

Requirement for the experiment and the thesis

For p_0 and Level 2C with a 109 min record, the inverse is Identifiable over the full tested range of peak frequency and peakedness at $L = 4096$. At $L = 2048$ it is Identifiable for 15 of 16 seas: a narrow sea peaked near 6 rad/s is Marginal on Welch bias. $L = 1024$ is not adequate for any sea. Stiffness precision is best when the energy lies toward the upper part of the retained band, but precision is not limiting at any tested shape (at most 0.091%). The sea's spectral shape matters through the estimator bias, which a longer Welch segment controls. **M1 hand-off:** $\omega_p = 6$ rad/s, $\gamma_J = 7$ (re-resolved from $\omega_p = 3.5$, $\gamma_J = 7$).

Output (Results/R2/)	Contents
R2_20261009_100148.mat / _summary.txt	results and config; headline numbers
R2_sweep.csv	48 rows ($16 \text{ seas} \times 3 L$): σ , d_W , κ , class, bins used
R2_L.comparison.csv, R2_M1.selection.csv	L verdicts and RMSE; M1 scores q
figures/R2_fig1 to R2_fig3 (.pdf/.png)	Figures 1 to 3
Superseded	R2_20261003_130557 (pre-amendment)